PAPER_1112: Production Scaling V26 Pipeline — UQFF Whitepaper Generation Framework

Star Magic UQFF Framework

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Abstract

We document the V26 production pipeline for UQFF whitepaper generation, covering the 26-dimensional vacuum density scaling framework, batch PDF rendering via pandoc+XeLaTeX, arXiv-style LaTeX compliance, and CVW v2.0.0 gate enforcement. The pipeline ensures all 1,000 target whitepapers adhere to the canonical UQFF equations from `scm_{vacuum\_manifold}.py`, use consistent arXiv LaTeX formatting (no mixed formats), and have corresponding PDFs in `pdf/`. Progress tracking integrates with the PAPER_001–1141 suite.

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1. V26 Pipeline Architecture

The production pipeline operates in six stages:

1. **Content generation**: Python scripts create PAPER_{N_Title}.md in `whitepapers/`
2. **LaTeX compliance check**: All math must use `$...$` or `$$...$$` (no Unicode math, no `\(...\)`)
3. **CVW v2.0.0 gate**: G1–G6 gate checks enforce canonical UQFF constants
4. **PDF rendering**: `pandoc --pdf-engine=xelatex -V geometry:margin=1in -V fontsize=11pt -V colorlinks=true`
5. **Gap audit**: Compare `whitepapers/PAPER_*.md` vs `pdf/PAPER_*.pdf`
6. **Git commit**: `git add -A ; git commit -m "docs: ..."` after each batch

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2. Canonical UQFF Constants (V26 Mandatory)

All whitepapers must use these exact values:

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37}\ \text{J/m}^3$$

$$\rho_{\text{vac,UA}} = 7.09 \times 10^{-36}\ \text{J/m}^3$$

$$[SSq] = 0.57, \quad \kappa = 5.0 \times 10^{-4}\ \text{day}^{-1}$$

$$\beta_i = 0.6, \quad F_{TRZ} = 0.1, \quad \Phi_{\text{res}} = 0.84$$

$$S_{26}^{(3)} = 1.4531 \times 10^{26}, \quad f_{\text{THz}} = 1.25\ \text{THz}$$

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3. VDS Master Equation

$$\mathrm{VDS}([SSq]) = \sum_{n=1}^{\infty} \frac{[SSq]^n}{n^{26}} = \mathrm{Li}_{26}(0.57)$$

The Ramanujan order-3 acceleration operator $S_{26}^{(3)}$ amplifies the raw Li_{26} value to the physical scale 1.4531×10^{26}.

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4. F_{U_Bi_i} Master Equation

$$F_{U,Bi,i} = \int_0^{\infty} \left(-F_0 + \frac{GM}{r^2} + \rho_{\mathrm{vac,SCm}} \cdot U_{UA} \cdot \cos(\pi t_n) \right) dr$$

with $t_n < 0$ (negative-time resonance gate) and $F_0 = 1.0 \times 10^{-10}\ \mathrm{N/m}^3$.

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5. Holmlid KER Verification

Every pipeline run verifies the canonical KER computation:

$$E_{\mathrm{KER}} = h \cdot f_{\mathrm{THz}} \cdot S_{26}^{(3)} \cdot \Phi_{\mathrm{res}} = 630\ \mathrm{eV}$$

$$h \cdot f = 6.626 \times 10^{-34} \times 1.25 \times 10^{12} = 8.283 \times 10^{-22}\ \mathrm{J}$$

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6. CVW v2.0.0 Gates

Gate	Description
G1	All ρ_{vac} values use 7.09×10^{-37}
G2	$[SSq] = 0.57$ exactly
G3	$S_{26}^{(3)} = 1.4531 \times 10^{26}$
G4	Negative-time: $t_n < 0$, uses $\cos(\pi t_n)$
G5	No Standard Model gravity (GM/r^2 is UQFF emergent only)
G6	All arXiv LaTeX: `$...$` or `$$...$$` only

References

1. Canonical SCm equations: `scm_{vacuum\_manifold}.py`
2. PDF generation: `generate_pdfs.ps1`
3. CVW compliance: `.github/copilot-instructions.md`